



Legacy Robotics

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Background

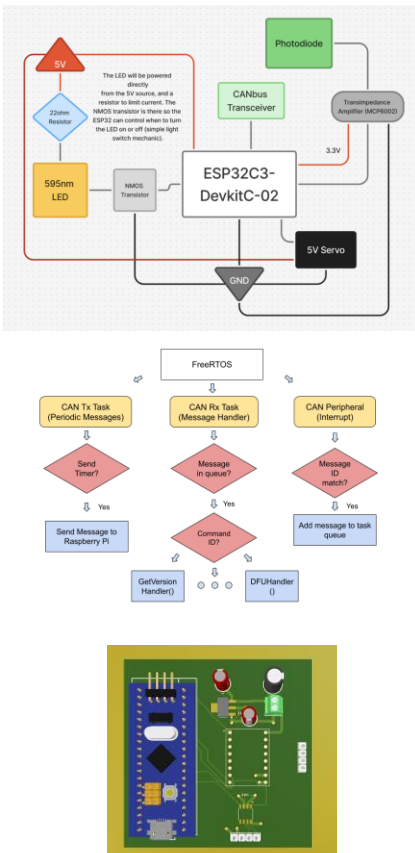
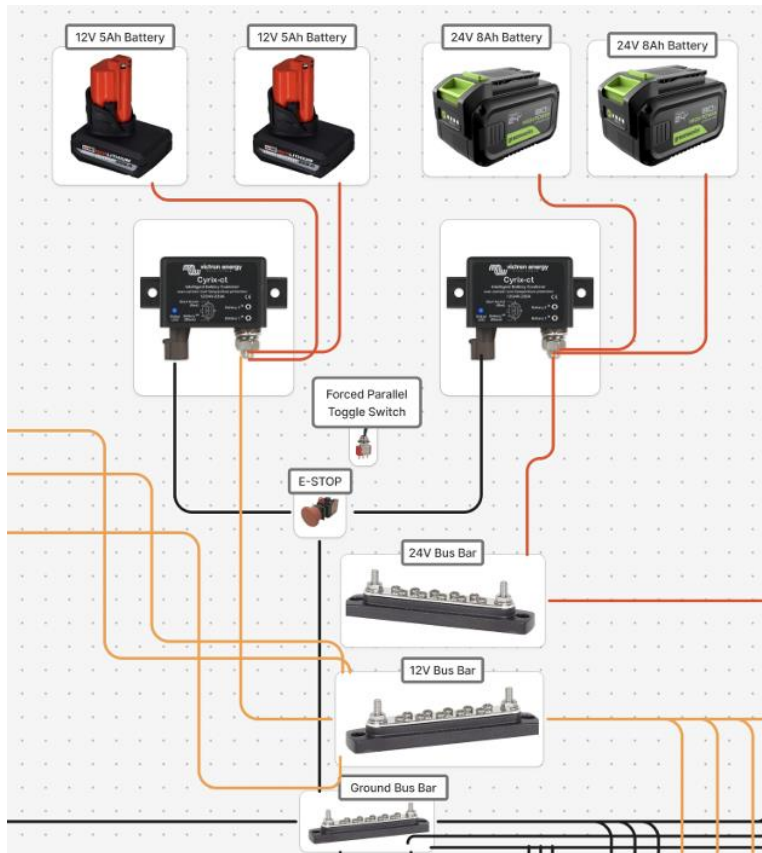
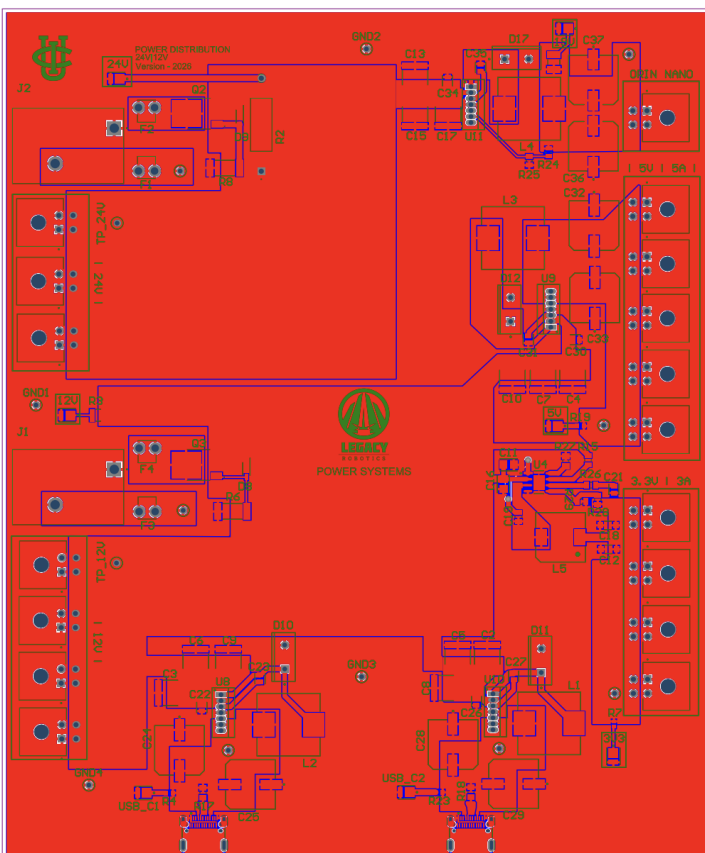
Legacy Robotics is a multidisciplinary team of 83 undergraduate and graduate students designing and building UC Irvine's first ever autonomous Mars rover (FESC) and drone for the University Rover Challenge, a global competition hosted annually by the Mars Society. Teams are tasked with developing the next generation of Mars rovers that will one day work alongside astronauts exploring the Red Planet. This year, with 116 universities and colleges representing 18 countries, the top 36 teams advance to the final competition held in Utah.

Objectives

- Deliver a fully-autonomous rover and drone capable of completing each of the 4 URC missions: Delivery, Science, Autonomous, and Equipment Servicing for the URC finals held May 27-30, 2026.
- Provide students who are interested in robotics and space exploration with an opportunity to showcase their skills in a globally recognized University level competition
- Receive nomination for the URC System Acceptance Review, a technical report & 5-minute video showing system capabilities.
- Continue building our **Legacy** for future generations of engineers to follow and build upon

Electrical Architecture

The **Electrical Team** oversees the rover's **power distribution, communication, and embedded systems**, ensuring efficient and reliable operation of all components. The electrical architecture is built around a robust power system, featuring a regulated distribution network that supplies multiple subsystems, including high-power motor drivers, embedded controllers, and scientific instruments. This includes designing **custom PCBs** for power management, integrating **RF communication** for long-range telemetry, and developing firmware for precise system control. By collaborating closely with other sub teams, we ensure seamless integration and full mission readiness.

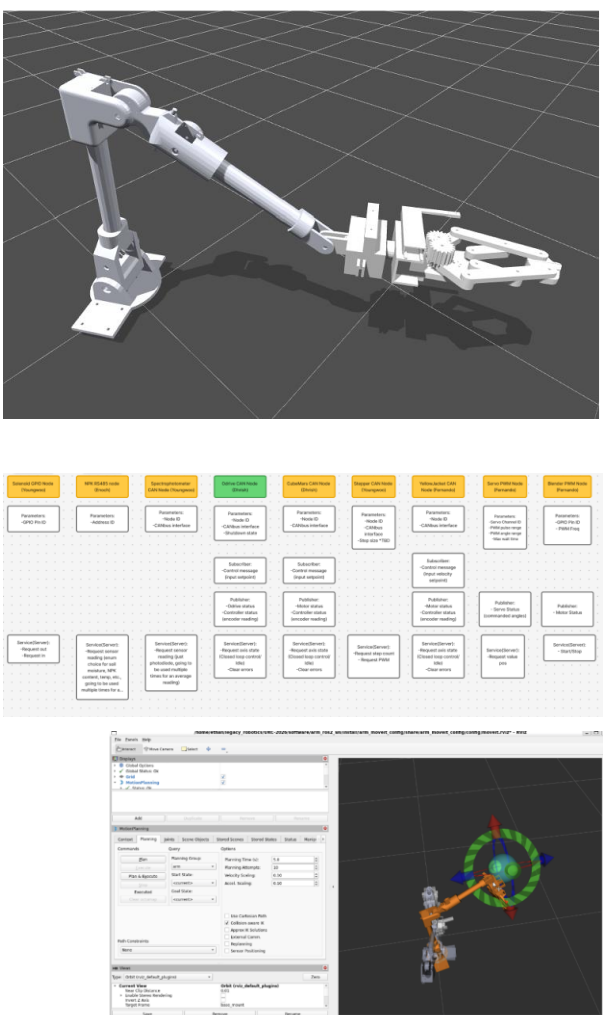
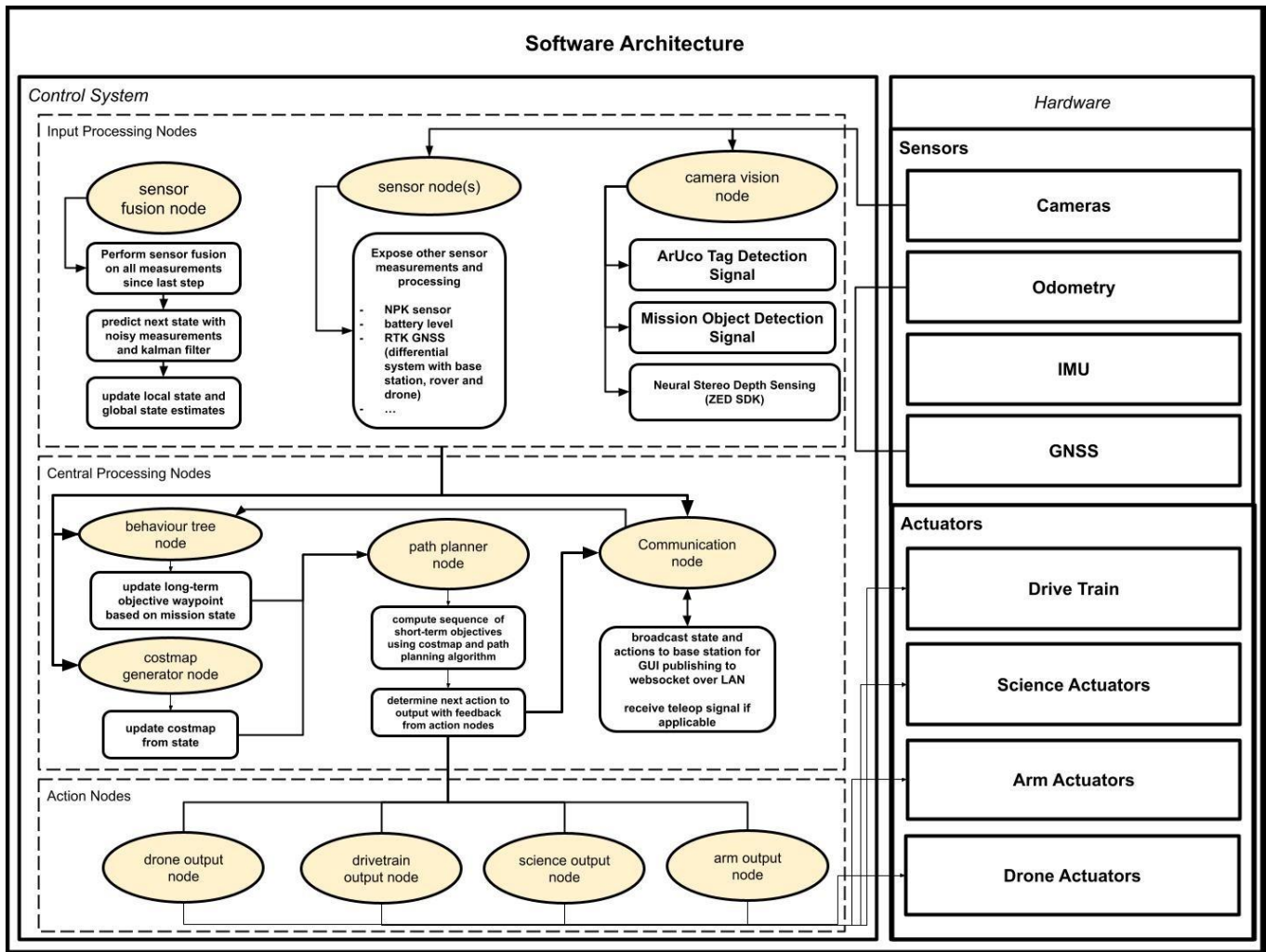


Assembled Rover (Arm Config.)

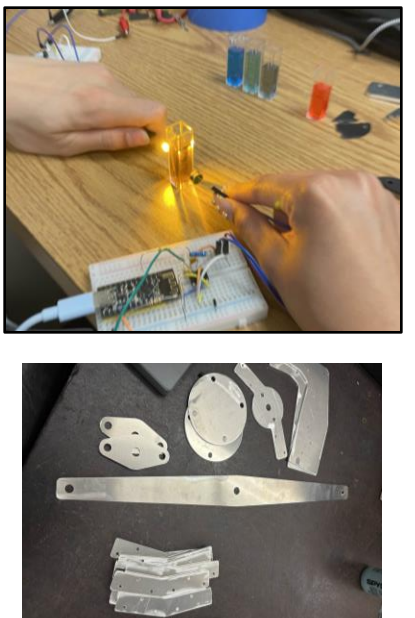
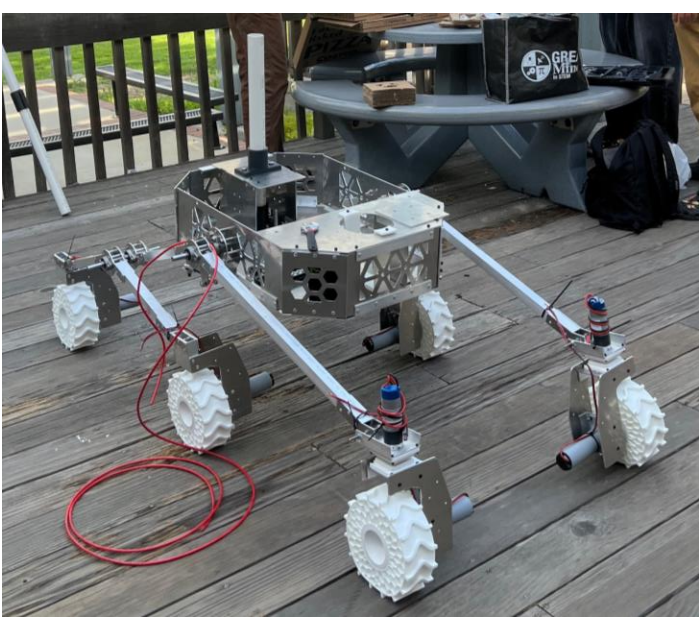
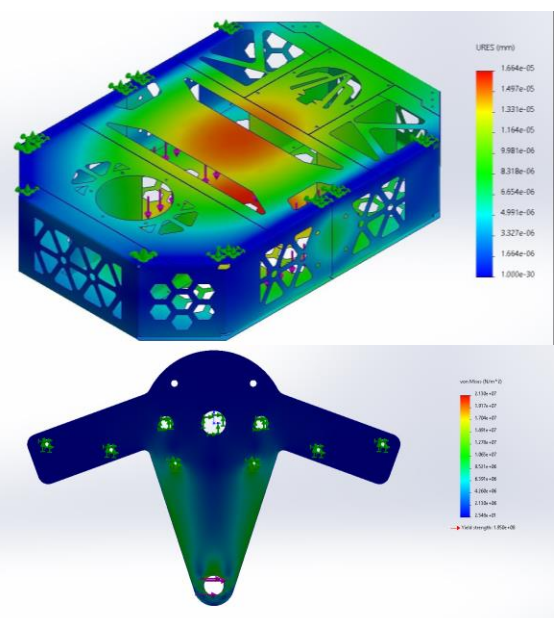


Software Architecture

The **Software Team** manages the movement of the rover's intricately designed subsystems, with the ultimate goal of achieving full autonomy. By utilizing **cameras, GNSS, IMU, and motor encoder data (ticks)**, both the rover and the team overseeing its operation can accurately determine its position relative to itself, its starting position, and its destination. Using this data, the rover autonomously navigates toward its goal, performs tasks such as object retrieval and drilling, and safely returns to the base station.



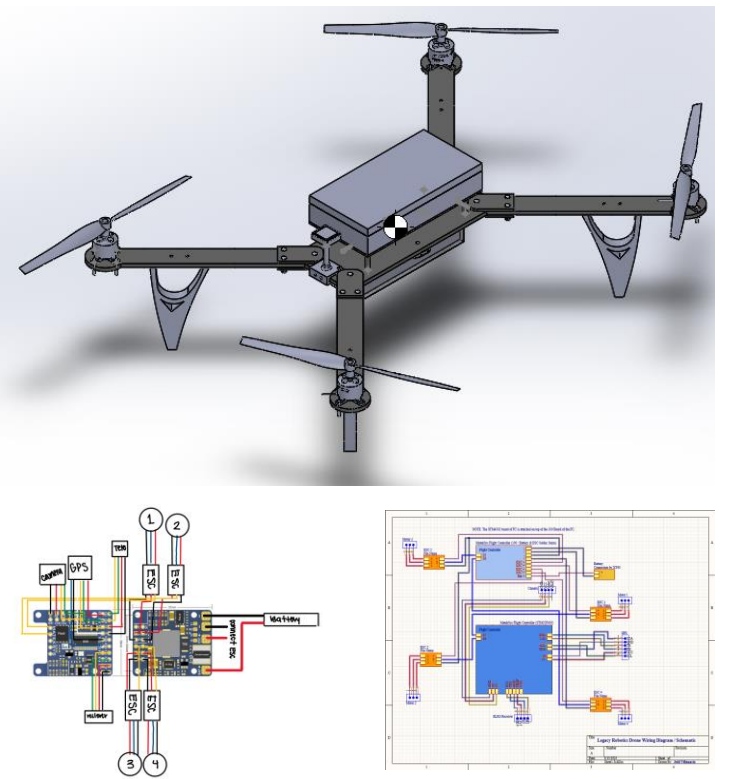
System Analysis & Prototype



Guidance Drone

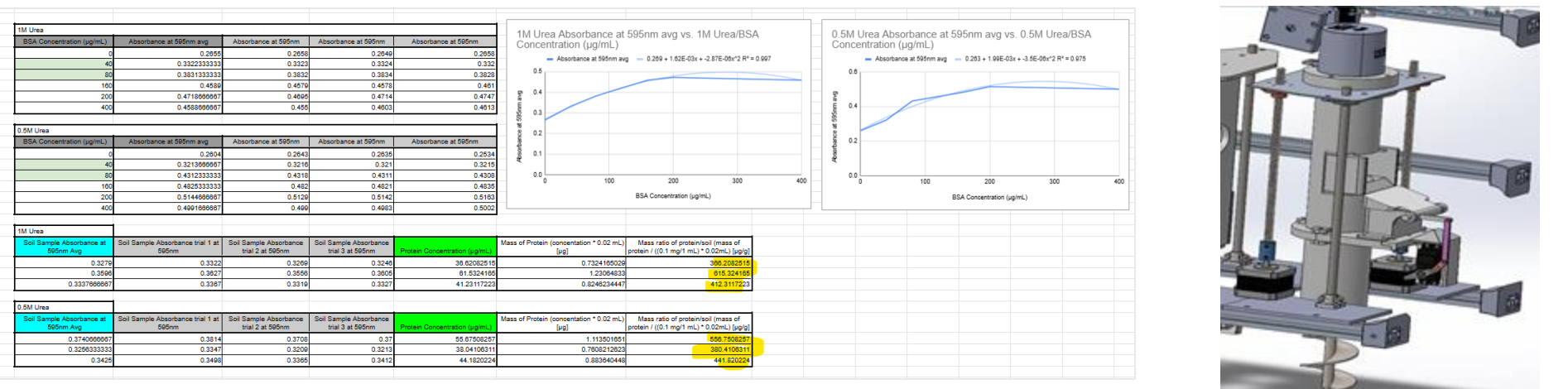
The guidance drone must act as a Another point of view/radio relay assisting in the Delivery Mission. Current Accomplishments include:

- Calculated 15-minute flight time
- Completed Test Board
- Each of the 4 motors spinning
- Tested sticks to ArduPilot
- Printed and tested frame and propellers



Science Payload

The **Science Payload** is a compact and efficient mechanical system designed to detect biosignatures in a Martian-analog environment. Developed and fabricated by the **Science Team**, the mechanism features an **auger drill** to collect subsurface soil and deposit it into a cache. A scooper is utilized to obtain additional soil for analysis via a modified **Bradford Assay**, a colorimetric method used to quantify soil protein content, performed with a custom **spectrophotometer**.



Project Sponsors

